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Cross-Channel Relational Knowledge Distillation for Low-Electrode SSVEP Brain–Computer Interfaces: A Multi-Backbone Reliability Study with Embedded Feasibility Assessment

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ABSTRACT

Reducing the number of electrodes in a steady-state visual evoked potential (SSVEP) brain-computer interface (BCI) improves wearability for applications such as wheelchair navigation and SSVEP-based spelling, but it degrades decoding accuracy. Cross-channel relational knowledge distillation (CCKD) recovers part of this loss by transferring the relational structure of a multi-electrode teacher to a low-electrode student. Our earlier work reported this gain for a single backbone, FB-tCNN, evaluated on one subject split. We extend that evaluation to five public backbones (FB-tCNN, DeepConvNet, ShallowConvNet, EEGNet, and Compact-CNN), two 40-target datasets (Benchmark and BETA), and two student montages, each with five independent subject splits and eight seeds per split. At a fixed distillation weight, 5 of 20 backbone-dataset-montage settings improved significantly and 4 degraded significantly before correction for multiple comparisons; only 2 settings, both on EEGNet with BETA, survived Holm-Bonferroni correction, and both were negative. Evaluation on a single subject split, the protocol used in our earlier work, misjudged the sign or significance of the effect in 10 of 20 settings. Swapping the student montage for an alternative electrode subset flipped the sign of the effect in 12 of 40 cases. The accuracy gain correlated with how much CCKD increased centered kernel alignment (CKA) between teacher and student features (Pearson $r = 0.55$, $p = 0.012$); CKA measured before distillation did not predict the gain ($r = -0.11$, $p = 0.65$). Distilling the last layer alone gave a larger gain than distilling an intermediate layer alone in only 5 of 20 settings; distilling both layers together matched or exceeded whichever single-layer choice was better in 8 of 20 settings. We further deploy the resulting student models on a Raspberry Pi 4 and a Xilinx ZCU104, the hardware classes used in wheelchair-control and SSVEP-speller prototypes, and report latency and post-quantization accuracy on both boards, PL synthesis feasibility and on-board accuracy for the smallest backbone, and board-level power on the ZCU104, where an on-board sensor is available (the Raspberry Pi 4 exposes no on-board power telemetry). These results show that cross-montage relational knowledge distillation is backbone- and montage-dependent, that single-split evaluation is unreliable at the effect sizes typical of this literature, and that the resulting low-electrode students remain feasible on both embedded targets at the latency, resource, and accuracy budgets measured.

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1. Introduction

A steady-state visual evoked potential (SSVEP) brain-computer interface (BCI) acts as the command-generation stage of a closed-loop control system: it senses the user's intent from EEG, maps the decoded target frequency to a discrete control command, and passes that command to a lower-level controller that actuates a wheelchair, a spelling interface, or a robotic arm [1, 2, 3, 4, 5]. As in any human-in-the-loop control architecture, the achievable command rate, measured in the BCI literature as the information transfer rate (ITR), sets the throughput of this channel, and the decision latency of the classifier must fit within the control loop's update period for the system to react safely to the

operator's intent [6, 7]. A multi-electrode wet-gel cap improves decoding accuracy and therefore the reliability of each generated command, but it also increases setup time and reduces wearability, which limits the use of such a controller outside the laboratory [8, 9, 10]. Reducing the electrode count to one or three channels improves wearability for a controller meant to be worn daily, but it degrades accuracy, because the classifier loses the spatial information carried by the discarded channels, which in turn raises the rate of missed or incorrect commands sent to the actuator [11, 12]. This trade-off between the wearability of the sensing front end and the reliability of the control commands it produces is the central problem addressed in this paper.

Knowledge distillation (KD) offers one way to narrow this gap: a teacher network trained on the full montage transfers information to a student network restricted to the low-electrode layout [13]. Cross-channel relational knowledge distillation (CCKD) is one such method. It preserves the pairwise similarity structure of teacher features, encoded as a Gram matrix, instead of matching feature values directly [14]. In earlier work, we showed that CCKD recovers part of the accuracy lost to montage reduction for one convolutional backbone, FB-tCNN, on two public 40-target SSVEP datasets [14, 15]. That evaluation, like most EEG knowledge-distillation studies, used a single backbone, a single distillation layer, and a single subject split [16, 17, 18, 19, 20, 21], leaving open whether the benefit generalizes to other architectures, to other electrode subsets, or to a more rigorous multi-split evaluation.

SSVEP decoding has moved from spatial-filter methods such as canonical correlation analysis toward a wide range of deep architectures, including filter-bank convolutional networks [15], transformer-based models [22, 23], graph convolutional networks [24], and attention-augmented compact networks [25, 26, 27, 28, 29]. These backbones differ in depth, receptive field, and the number of internal feature stages at which a teacher-student match can be imposed, so a distillation strategy tuned for one architecture need not transfer to another. A separate line of work shows that EEG machine-learning results are sensitive to how data are split and how significance is assessed: segment-level or single-split evaluation can invert the sign of an effect or inflate its apparent size [30, 31, 32, 33, 34, 35, 36]. Distillation gains reported in the sub-percentage-point range are exactly the regime in which this sensitivity matters most, yet our earlier CCKD evaluation relied on a single subject split [14].

Centered kernel alignment (CKA) quantifies how similar two sets of neural-network representations are, independent of rotation and scale, and is a standard tool for comparing layers across trained networks [37, 38, 39, 40]. It matches the quantity that CCKD already optimizes, because the training objective matches the Gram matrix of pairwise sample similarities between teacher and student. CKA-based regularizers have been applied directly inside EEG backbones [41], but using the *change* in CKA induced by distillation to predict how much accuracy the distillation step will recover remains untested in SSVEP decoding. Finally, deploying any of these backbones on the embedded hardware used in real wheelchair and speller prototypes, such as a Raspberry Pi 4 or a Xilinx FPGA, requires quantization and resource-aware implementation that can itself change classification accuracy [42, 43, 44, 45]. Whether a knowledge-distilled low-electrode model keeps its accuracy advantage after this step is a question the SSVEP-BCI literature has not addressed directly.

This paper extends our single-backbone conference evaluation [14] along four directions to close these gaps. The contributions of this research are as follows.

(i) A controlled evaluation of CCKD across five public backbones (FB-tCNN, DeepConvNet, ShallowConvNet, EEGNet, and Compact-CNN), two 40-target datasets, and two low-electrode montages, each with five independent subject splits and eight seeds per split, with Holm-Bonferroni correction for multiple comparisons.

(ii) An analysis of centered kernel alignment between teacher and student features as a predictor of distillation benefit, including an intervention that distills an intermediate layer instead of the

penultimate layer used in prior work.

(iii) A direct comparison of single-split and multi-split evaluation on the same data, quantifying how often single-split evaluation misjudges the sign or significance of the effect.

(iv) A deployment of the resulting student models on a Raspberry Pi 4 and a Xilinx ZCU104, the hardware classes used in wheelchair-control and SSVEP-speller prototypes, reporting decision latency against the control-loop update period those prototypes require, post-quantization accuracy, PL synthesis feasibility and on-board accuracy for the smallest backbone, and board-level power on the ZCU104 (the Raspberry Pi 4 has no on-board power sensor).

2. Related Work

2.1. Deep Learning Backbones for SSVEP Decoding

Classical SSVEP recognition relies on frequency-domain correlation methods, such as canonical correlation analysis and task-related component analysis, which compare the recorded signal against a reference template at each stimulation frequency. Convolutional and attention-based networks have since replaced or augmented these templates, achieving higher accuracy with less calibration data. FB-tCNN combines a filter bank with a time-domain convolutional network to handle short time windows [15] and is the backbone our 2026 conference paper distilled into a low-electrode student [14]. ConvCA replaces the fixed correlation template with a learned convolutional correlation layer [46]. SSVEPformer and related transformer architectures model long-range dependencies in the frequency-domain spectrum rather than the raw time series [22, 23]. Graph-convolutional variants such as DDGCNN exploit the spatial layout of electrodes directly [24], and lightweight attention networks such as the task-attention model of Wang et al. and LGFCNN reduce the number of trainable parameters needed for individual calibration [26, 28]. Several studies address the few-channel regime directly, either through specialized architectures [47, 48, 12] or through analyses of how channel count and placement affect accuracy [11, 49]. Across this body of work, each architecture is typically evaluated and tuned on its own; comparative multi-backbone studies of any post-hoc transfer technique, including knowledge distillation, are rare.

2.2. Knowledge Distillation for EEG and Brain-Computer Interfaces

Knowledge distillation transfers information from a larger teacher network to a smaller or more constrained student network, usually by matching output logits, intermediate features, or pairwise feature relations. In EEG-based BCIs, similarity-keeping distillation reduces the channel count of a classifier while preserving most of its accuracy [13], which is the closest precedent to the cross-channel relational objective evaluated in this paper. Other EEG and physiological-signal studies use distillation mainly across subjects or domains rather than across channel counts: dual-teacher and multi-teacher frameworks combine several source models for motor-imagery decoding [17, 50], teacher-assistant chains ease a large capacity gap between teacher and student [21], and self-distillation or domain-generalization objectives target cross-subject and cross-dataset emotion recognition [18, 20, 51, 52]. A cross-modal variant distills knowledge from a visual encoder into an EEG encoder for continuous emotion recognition [19]. One study combines distillation with quantization-aware training in a single objective for EEG-based emotion recognition [53], which anticipates the combined knowledge-distillation-and-embedded-deployment pipeline evaluated in this paper. None of these studies evaluate whether a channel-reduction distillation objective generalizes across backbone architecture, which is the first research question addressed here.

2.3. SSVEP-Based Control Applications: Wheelchairs, Spellers, and Robotic Arms

In an SSVEP-controlled assistive device, the decoder is one stage in a closed control loop: the stimulus interface presents a fixed set of flicker targets, the decoder maps the detected frequency to one discrete command out of that set, a low-level controller converts the command into a motion or selection primitive, and the device's motion or the speller's screen feedback closes the loop back to the operator. The number of commands available per unit time, and therefore the number of independent degrees of freedom the operator can control directly, is set jointly by the number of stimulus targets and the decoder's ITR [6, 7]. Wheelchair-control systems built on this loop use SSVEP alone for discrete heading commands [1, 2], SSVEP combined with electrooculography [54] or P300 potentials [3] to add command channels, and SSVEP combined with computer vision and augmented reality, where the onboard perception system supplies continuous obstacle avoidance and the SSVEP channel supplies only the high-level heading command, reducing the decoding accuracy and update rate the BCI stage must guarantee for safe operation [3, 4]. Systematic reviews of this application area report that most prototypes still use eight or more electrodes, in part because the stimulation and classification pipeline was developed for a research-grade cap rather than a long-wear consumer headset, and that few report a closed-loop latency budget explicitly [55, 56]. Speller interfaces face the same loop with a much larger command set: encoding more target characters needs a larger stimulus set and a correspondingly higher perceptual workload per selection, which motivates paradigm designs that enhance the discriminability of each frequency and keep the per-character decision time bounded, such as frequency-enhanced row-column spellers [6] and adaptive spatial filtering [7]. Robotic-arm control shares the same channel-count trade-off at a finer grain: reducing the montage improves comfort for extended operation, but it narrows the margin between target frequencies that the decoder can resolve reliably, which directly limits how many independent joint or end-effector commands the control loop can safely accept [5]. These applications motivate the two deployment scenarios evaluated later in this paper: a three-electrode montage intended for a wheelchair headset, and a comparison across electrode subsets relevant to speller and robotic-arm command sets.

2.4. Representation Similarity and Centered Kernel Alignment

Centered kernel alignment (CKA) measures how similar two sets of representations are, without requiring the same number of units or a fixed correspondence between them. The Gaussian and linear variants of this measure have become standard tools for comparing layers across trained networks [37, 38, 39]. Kriegeskorte and Wei situate CKA within the broader framework of representational geometry, which compares the relative distances or similarities among stimuli rather than their absolute feature values [40]. This framework matches the quantity CCKD already optimizes: the training objective matches the Gram matrix of pairwise sample similarities between teacher and student, which is the same object CKA uses to quantify representational similarity. Tobon-Henao et al. apply a centered-alignment regularizer directly inside an EEGNet classifier for motor-imagery discrimination [41], confirming that CKA-based objectives are compatible with EEG backbones. To our knowledge, no prior SSVEP study has used the change in CKA induced by distillation as a predictor of how much accuracy the distillation step recovers, which is the second research question addressed here.

2.5. Statistical Reliability in Cross-Subject EEG Evaluation

EEG classification accuracy is sensitive to how training, validation, and test data are partitioned across subjects and sessions. White and Power show that k-fold cross-validation without subject-level separation substantially overestimates accuracy in passive BCI designs [30]. Del Pup et al. and Kamrud et al. report that subject-wise partitioning choices alone change measured performance by several percentage points in cross-subject deep-learning models [31, 34]. Brookshire et al., Lee et al., and Geethapriya and Pattabiraman document the same leakage pattern in translational and clinical EEG classifiers, where segment-level splitting lets a classifier exploit subject identity instead of the

intended signal [32, 33, 36]. Xu et al. show that accuracy estimates also vary across different EEG datasets collected under nominally the same paradigm, independent of any single-subject leakage [35]. These findings motivate evaluating CCKD on five independent subject splits with Holm-Bonferroni correction, rather than on the single split used in the original conference evaluation, and motivate the direct single-split-versus-multi-split comparison reported as the third research question of this paper.

2.6. Quantization and Embedded Deployment for Real-Time BCIs

In a control loop, the decoder's execution time is part of the loop's dead time: it adds directly to the delay between the operator's intent and the actuator's response, so an embedded implementation must keep that delay within the margin the mechanical system can tolerate without instability or a perceptibly sluggish response. Deploying a trained SSVEP classifier on embedded hardware for this kind of real-time, untethered operation typically requires quantization, pruning, or a dedicated hardware accelerator, because a floating-point convolutional network exceeds the memory and power budget of a wearable or battery-powered controller. Pacini et al. compare CPU, GPU, and FPGA deployments of a motor-imagery CNN and report large differences in latency and power across platforms for the same trained model, which shows that the choice of compute platform, not only the model, sets the achievable control-loop rate [43]. Novac et al. provide a general methodology for quantizing and deploying deep networks on microcontrollers [45], and Palumbo et al. demonstrate a full FPGA-embedded BCI pipeline for locked-in patients, with the decoder running on the same board that generates the output command [44]. Spiking and event-driven architectures offer an alternative low-power path for wearable physiological-signal classification, demonstrated for ECG and seizure detection on FPGA [57, 58, 59, 60, 61, 62], and a recent survey catalogs edge-AI accelerator designs for brain-computer interfaces specifically [42, 63]. None of this work evaluates whether a knowledge-distilled student model, already trained to approximate a larger teacher, retains its accuracy advantage after the additional accuracy loss introduced by embedded quantization, or whether its latency on this class of hardware fits the control-loop budget required by a wheelchair or speller controller, which motivates the fourth research question addressed in this paper.

2.7. Summary and Research Gap

Across these five areas, three gaps motivate the present study, all bearing on the same practical question: can a low-electrode, knowledge-distilled SSVEP decoder serve as the command-generation stage of a real wheelchair, speller, or robotic-arm control loop without an unacceptable loss of command reliability or an unacceptable increase in decision latency? First, cross-channel knowledge distillation for SSVEP decoding has been evaluated on a single backbone, so its generalization across architecture, and therefore its generalization across the different embedded platforms that implement each architecture, is unknown. Second, no predictor of distillation benefit has been tested for this setting, even though centered kernel alignment is readily available for this purpose and could guide which backbone-montage pair to deploy in a given control application. Third, the methodological risks documented for EEG cross-subject evaluation, including single-split overestimation, have not been applied to knowledge-distillation studies specifically, and the embedded feasibility of the resulting student models, including whether their decision latency fits a realistic control-loop budget, has not been measured directly. The remaining sections of this paper address each gap in turn: Section 3 describes the backbones, datasets, and distillation protocol; Section 4 reports, in sequence, the multi-backbone and montage results, the CKA analysis and the layer-intervention experiment, the single-split-versus-multi-split comparison, and the embedded deployment results, relating decoder latency and accuracy to the control-loop requirements of the wheelchair and speller use cases introduced above; Section 5 concludes.

3. Method

3.1. CCKD

The teacher receives input from nine parieto-occipital electrodes, is pretrained with cross-entropy, and is then kept fixed. The student shares the teacher's architecture but receives only three or one electrode. For a mini-batch of N samples, let $F_T^{(l)} \in \mathbb{R}^{N\times d_T}$ and $F_S^{(l)} \in \mathbb{R}^{N\times d_S}$ denote the teacher and student features at layer l. The Gram matrix of each model is $G^{(l)} = F^{(l)}(F^{(l)})^\top \in \mathbb{R}^{N\times N}$; because the Gram matrix always has size $N \times N$, CCKD does not require $d_T = d_S$. Each row is normalized by its ℓ_2 norm, $\hat{G}^{(l)}[i,:] = G^{(l)}[i,:]/\|G^{(l)}[i,:]\|_2$, and the relational loss at layer l is

$$\mathcal{L}_{\text{CCKD}}^{(l)} = \frac{1}{N^2}\left\|\hat{G}_T^{(l)} - \hat{G}_S^{(l)}\right\|_F^2. \tag{1}$$

For a set of distillation layers $\mathcal{K}$, the student optimizes $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_{sp}\,|\mathcal{K}|^{-1}\sum_{l\in\mathcal{K}}\mathcal{L}_{\text{CCKD}}^{(l)}$, where $\mathcal{L}_{\text{CE}}$ is the student's cross-entropy loss and λ_{sp} is the distillation weight. After training, the teacher is discarded, so CCKD does not change the student's architecture, parameter count, or inference cost [14].

In [14], FB-tCNN was distilled at two layers, while the other architectures were distilled only at the last layer, so architecture and distillation-layer count were confounded. This paper defines the same two layers for every backbone: the **intermediate layer** (`mid`, the output of the spatial convolution block immediately after batch normalization and the nonlinearity, whose width does not depend on the number of input electrodes) and the **last layer** (`last`, the penultimate feature, the classifier's input). The main protocol distills only the last layer ($\mathcal{K} = \{\texttt{last}\}$); the intervention experiment in Section 4.3 additionally compares $\mathcal{K} = \{\texttt{mid}\}$ and $\mathcal{K} = \{\texttt{mid}, \texttt{last}\}$.

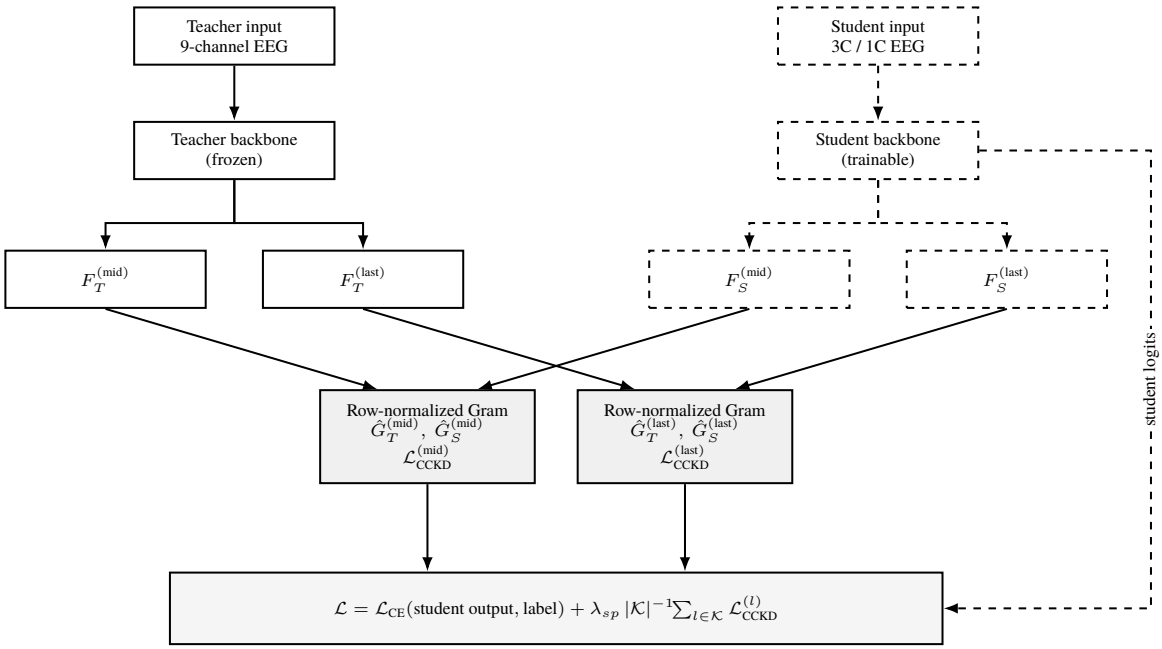


Fig. 1. Block diagram of CCKD. The fixed teacher (blue, solid) and the trainable student (green, dashed) both output features at the `mid` and `last` layers; each layer computes a row-normalized Gram matrix, which is then matched by $\mathcal{L}_{\text{CCKD}}$ (red). The total loss $\mathcal{L}$ adds the student's cross-entropy loss.

3.2. Five Backbones

Table 1 lists the five public backbones used in the evaluation: FB-tCNN [15], DeepConvNet and ShallowConvNet [64], EEGNet [65], and Compact-CNN [66]. A few hyperparameters were adjusted for the 1.0 s window (250 samples at 250 Hz): DeepConvNet keeps three convolution–pooling blocks instead of four; ShallowConvNet uses pooling of 35 samples with stride 7; Compact-CNN uses a temporal kernel of 64 instead of 256.

Table 1. The five backbones and the architectural properties used as explanatory variables in RQ2.

Backbone	Spatial convolution	Nonlinearity	Max. temporal kernel
FB-tCNN	ordinary, shared across 3 filter bands	ELU	250
DeepConvNet	ordinary	ELU	10
ShallowConvNet	ordinary	square, log	25
EEGNet	depthwise	ELU	64
Compact-CNN	depthwise	ELU	64

3.3. Datasets and Montages

Two public 40-target SSVEP datasets, sampled at 250 Hz, are used: Benchmark, with 35 subjects, and BETA, with 70 subjects. Each subject split assigns disjoint subjects to the training, validation, and test sets (Benchmark: 21/4/10; BETA: 42/8/20). The test set is evaluated only once. Five splits use split seeds 2026, 3026, 4026, 5026, and 6026; each split runs eight model-initialization seeds (2026–2033).

The teacher uses nine electrodes: Pz, PO5, PO3, POz, PO4, PO6, O1, Oz, O2. The student's original montage is {O1, Oz, O2} (3C) and {Oz} (1C) [14]. To answer RQ1b on montage sensitivity, this paper adds two three-electrode montages, {PO3, POz, PO4} (3C_PPO) and {POz, O1, O2} (3C_POO), and two one-electrode montages, {POz} (1C_POz) and {O1} (1C_O1), fixed before running the experiments.

3.4. Training and Validation Protocol

Every model is trained for 100 epochs with Adam (learning rate 10^{-3}, weight decay 10^{-4}), a cosine-annealing schedule, and a batch size of 64; the checkpoint is selected by subject-averaged validation accuracy. The primary metric is classification accuracy on the test subjects. The gain Δ is the paired difference between the distilled student and the baseline student (trained with cross-entropy only) within the same run, expressed in percentage points (pp), averaged over seeds and then over subjects.

For RQ1 (a fair comparison across backbones), $\lambda_{sp} = 100$ is held fixed for every backbone, using the same unified last-layer distillation protocol from Section 3.1, unlike the earlier protocol, in which FB-tCNN and the other backbones did not use the same number of distillation layers. For RQ1b, RQ2, and the layer-intervention experiment, λ_{sp} is chosen separately for each (dataset, backbone) pair by majority vote of the best validation λ_{sp} across the five splits (Table 2), so that each backbone is evaluated at a weight close to its own optimum rather than being forced to use $\lambda_{sp} = 100$ when that value does not suit it.

On a single split, statistical significance is tested with the Wilcoxon signed-rank test across subjects. Across five splits, significance is tested with a one-sample t-test on the five Δ values (H_0: $\bar{\Delta} = 0$). When multiple tests belong to the same family (the 20 backbone–dataset–montage combina-

Table 2. λ_{sp} chosen by majority vote on validation (5 splits), used for RQ1b, RQ2, and the layer-intervention experiment.

Dataset	FB-tCNN	DeepConvNet	EEGNet	ShallowConvNet	Compact-CNN
Benchmark	100	100	10	10	10
BETA	100	1000	10	100	10

tions in RQ1), p-values are corrected with the Holm–Bonferroni procedure [67].

4. Result and Discussion

4.1. RQ1: Gain by Backbone at a Fixed Weight

Table 3 summarizes the gain from CCKD across five backbones, two datasets, and two montages, at a fixed $\lambda_{sp} = 100$ and the unified last-layer distillation protocol for every backbone (20 settings, each with 5 splits $\times$ 8 seeds).

Table 3. CCKD gain over 5 splits ($\lambda_{sp} = 100$ for every backbone, last-layer distillation). Each cell: $\bar{\Delta}$ (pp) and p from the one-sample t-test across splits. **Bold**: $p < 0.05$ (uncorrected). †: still significant after Holm–Bonferroni correction over all 20 tests ($p_{\text{Holm}} < 0.05$).

Backbone	Benchmark 3C	Benchmark 1C	BETA 3C	BETA 1C
FB-tCNN	**+0.78 (0.0043)**	+0.25 (0.138)	**+1.09 (0.0075)**	−0.15 (0.658)
DeepConvNet	+0.36 (0.050)	**+0.14 (0.025)**	**+0.49 (0.023)**	+0.60 (0.065)
EEGNet	+0.98 (0.094)	−0.36 (0.310)	−1.33 **(0.0008)**†	−1.33 **(0.0017)**†
ShallowConvNet	−0.75 **(0.043)**	−1.01 **(0.011)**	+0.44 (0.278)	**+0.49 (0.034)**
Compact-CNN	+0.18 (0.646)	−0.43 (0.124)	−0.16 (0.485)	−0.15 (0.221)

Of the 20 settings, 5 improved significantly and 4 degraded significantly before correction; after Holm–Bonferroni correction, only 2 settings remained significant, and both were negative (EEGNet on BETA, both 3C and 1C). The results do not rank by architecture family: FB-tCNN and DeepConvNet tend to improve (positive in 3/4 and 3/4 settings), ShallowConvNet degrades on Benchmark but improves on BETA, and EEGNet flips sign between the two datasets (improving on Benchmark, degrading strongly and significantly on BETA). This shows that the earlier conclusion that CCKD improves FB-tCNN [14] does not generalize to CCKD in general; the gain depends on both the backbone and the dataset.

4.2. RQ1b: Sensitivity to Montage

Table 4 compares the gain at the original montage with two alternative montages for each 3C/1C setting, at λ_{sp} chosen according to Table 2 (40 alternative settings).

Swapping the student montage for a different electrode subset flipped the sign of the effect in 12 of 40 cases. The sign flips occur in both directions (from beneficial to not beneficial and the reverse) and are not concentrated in one backbone or one dataset, although ShallowConvNet and Compact-CNN show a higher flip rate. This shows that the benefit of CCKD is sensitive to electrode placement, not only to electrode count.

Table 4. Gain ($\bar{\Delta}$, pp, over 5 splits) at the original montage compared with two alternative montages per setting. *: sign flips relative to the original montage.

	3 electrodes			1 electrode		
Backbone	Original	PPO	POO	Original	POz	O1
Benchmark						
FB-tCNN	+0.78	+0.24	+0.70	+0.25	$-0.09^{\star}$	+0.83
DeepConvNet	+0.36	+0.20	+0.02	+0.14	+0.20	+0.81
EEGNet	+1.60	+1.14	+0.64	+0.51	+0.92	$-0.06^{\star}$
ShallowConvNet	-0.15	$+0.09^{\star}$	-0.18	-0.42	-0.30	$+0.05^{\star}$
Compact-CNN	+0.26	$-0.11^{\star}$	+0.31	-0.17	$+0.29^{\star}$	$+0.04^{\star}$
BETA						
FB-tCNN	+1.09	+0.54	+1.10	-0.15	$+0.17^{\star}$	$+0.43^{\star}$
DeepConvNet	+0.93	+1.34	+0.80	+1.12	+1.10	+1.68
EEGNet	+0.11	+0.24	+0.02	+0.03	$-0.17^{\star}$	+0.18
ShallowConvNet	+0.44	+0.65	+0.13	+0.49	+0.04	+0.61
Compact-CNN	+0.01	$-0.03^{\star}$	+0.14	+0.16	$-0.00^{\star}$	+0.03

4.3. RQ2: Predictors of the Gain

4.3.1 CKA Similarity Between Teacher and Student

The loss (1) pulls the student's Gram matrix toward the teacher's Gram matrix; if CCKD is useful through this mechanism, the accuracy gain must go together with an increase in CKA. Linear CKA is measured between the teacher's and student's last-layer features on the test set, at λ_{sp} according to Table 2. ΔCKA denotes the CKA of the distilled student minus the CKA of the baseline student, averaged over 5 splits and 8 seeds.

Across the 20 settings, ΔCKA correlates with $\bar{\Delta}$ (Pearson $r = 0.55$, $p = 0.012$; Spearman $\rho = 0.57$, $p = 0.009$), see Figure 2. The baseline student's CKA (before distillation) does not correlate with the gain (Pearson $r = -0.11$, $p = 0.65$; Spearman $\rho = -0.03$, $p = 0.91$). This shows that the initial representational gap between teacher and student does not predict the effectiveness of CCKD; what predicts it is *how much CCKD narrows that gap*, consistent with the mechanism of the loss function.

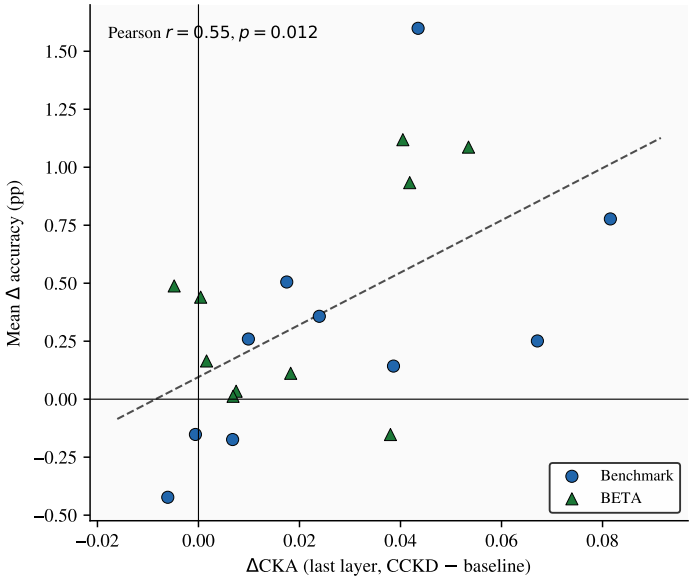


Fig. 2. Last-layer CKA increase induced by CCKD versus accuracy gain, for 20 backbone–dataset–montage settings. Dashed line: linear regression. Circles: Benchmark; triangles: BETA.

4.3.2 Distillation-Layer Intervention Experiment

The correlation above cannot distinguish two explanations: that the CKA increase causes the gain, or that both are caused by some other architectural property. To separate these possibilities, we vary the set of distillation layers $\mathcal{K} \in \{\texttt{last}, \texttt{mid}, \{\texttt{mid}, \texttt{last}\}\}$ while holding the backbone, dataset, montage, and λ_{sp} fixed.

Across the 20 settings, distilling the last layer alone gave a larger gain than distilling the intermediate layer alone in only 5 settings; distilling both layers together matched or exceeded whichever single-layer choice was better in 8 settings. In other words, the main protocol (last-layer distillation [14]) is not the best choice in most settings – distilling the intermediate layer, alone or combined with the last layer, usually performs as well or better. The per-setting accuracy and ΔCKA at each distilled layer are available in the code repository accompanying this paper (see the Reproducibility section).

4.4. RQ3: Reliability of Single-Split versus Multi-Split Evaluation

Table 5 compares the conclusion drawn from a single split (2026, Wilcoxon test across subjects) with the conclusion from five splits (t-test across splits), for the 20 settings at $\lambda_{sp} = 100$ (the RQ1 protocol).

Single-split evaluation misjudges the sign or statistical significance of the effect in 10 of 20 settings – exactly half. The discrepancies go in both directions: a single split can miss a stable improvement or degradation (for example, FB-tCNN Benchmark-3C, DeepConvNet 1C on both datasets), or it can report an effect that does not replicate under multiple splits (for example, ShallowConvNet BETA-3C). The two tests do not share the same unit of replication (Wilcoxon replicates across subjects, the t-test replicates across splits), so the table does not indicate which test has higher statistical power; what the table shows is that a study reporting only a single split can reach a conclusion opposite to the one reached when repeating the evaluation on different subject sets, at the effect sizes typical of the SSVEP CCKD literature (under 1–2 pp).

Table 5. Conclusion from a single split (2026) versus five splits, $\lambda_{sp} = 100$. $+/-/0$: improves/degrades/not significant ($p < 0.05$). *: conclusions differ (in sign or statistical significance).

Backbone	Student	Single split (pp, p)	Five splits (pp, p)	Benchmark	BETA
FB-tCNN	3C	$+0.31$ (0.359)	$+0.78$ (0.0043)	$0 \to +^\star$	$0 \to +^\star$
	1C	$+0.22$ (0.447)	$+0.25$ (0.138)	$0 \to 0$	$0 \to 0$
DeepConvNet	3C	$+0.17$ (0.492)	$+0.36$ (0.050)	$0 \to 0$	$+ \to +$
	1C	$+0.01$ (0.980)	$+0.14$ (0.025)	$0 \to +^\star$	$0 \to 0^\star$
EEGNet	3C	-0.20 (0.770)	$+0.98$ (0.094)	$0 \to 0^\star$	$- \to -$
	1C	$+0.32$ (0.557)	-0.36 (0.310)	$0 \to 0^\star$	$- \to -$
ShallowConvNet	3C	-0.52 (0.322)	-0.75 (0.043)	$0 \to -^\star$	$0 \to 0$
	1C	-0.67 (0.041)	-1.01 (0.011)	$- \to -$	$+ \to +$
Compact-CNN	3C	-0.15 (0.820)	$+0.18$ (0.646)	$0 \to 0^\star$	$0 \to 0^\star$
	1C	-0.97 (0.014)	-0.43 (0.124)	$- \to 0^\star$	$0 \to 0$

4.5. RQ4: Embedded Deployment

We apply static post-training quantization (PTQ) to INT8 to the student models already trained for RQ1, re-evaluating only on the test set without retraining. For the 2 backbone–dataset–montage settings that remained statistically significant after Holm–Bonferroni correction (RQ1, both on EEGNet with BETA, montages 3C and 1C), the accuracy deviation caused by PTQ kept the same sign as FP32 in all 10/10 split–seed settings (100%) and did not exceed 0.26 pp, within the 0.3–0.5 pp background-noise range measured in RQ3.

We further sweep the fixed-point bit width $\{16, 12, 10, 8\}$ using hls4ml's bit-accurate simulation for EEGNet and DeepConvNet, the smallest and largest backbones by actual parameter count (EEGNet: 5,640–5,672 parameters; DeepConvNet: 99,940–101,190 parameters, depending on montage, versus 44,488–90,536 for the other three backbones), on the BETA dataset with 5 splits. EEGNet's accuracy stays within 1 pp of FP32 at 10 bits or above and drops by more than 4 pp below 8 bits at both montages, matching the 8–10-bit breakpoint range predicted before measurement. DeepConvNet (montage 3C) remains nearly unchanged at 10–16 bits (deviation under 0.05 pp) but drops sharply below 8 bits (deviation of 1.5 pp at 8 bits, 20.4 pp at 6 bits, with an additional sweep around the breakpoint within the range locked in advance), showing that the true breakpoint lies between 6 and 8 bits, one step lower than the predicted range.

We measure the inference latency of all 5 backbones at FP32 and INT8 on a Raspberry Pi 4 (Cortex-A72) and the Cortex-A53 processing system (PS) of a Xilinx ZCU104, each setting averaged over 7 independent processes, 100 inferences per process, batch size 1, 1.0 s window. Here, INT8 uses weight-only dynamic quantization (only the convolution/linear layer weights are quantized; activations remain at FP32): static quantization raised an incompatibility error with the quantized kernel for the ELU and square operators used in FB-tCNN, DeepConvNet, and ShallowConvNet, confirmed identically on both boards; EEGNet and Compact-CNN also use ELU but combine it with depthwise rather than ordinary spatial convolution and did not hit this error, so ordinary spatial convolution, not the nonlinearity alone, is the more likely distinguishing factor. At INT8 width, all 5 backbones have a forward-pass latency under 16 ms on both boards, including the added filter-bank time for FB-tCNN (maximum total pipeline of 24.5 ms on the ZCU104 PS). This compute-only latency is about 12 times below an illustrative 300 ms control-loop budget that is assumed rather than sourced from a specific wheelchair or speller controller, and it excludes the 1.0 s SSVEP stimulation window that the same protocol requires before this segment becomes available, so it bounds only the classifier's own contribution to the control loop's dead time, not the full decision-to-actuation latency. On the ZCU104 PS, board power measured via the on-board INA226 sensor rises from an idle baseline of 11.75 W to

12.32–12.45 W during continuous back-to-back inference, an inference-attributable increase of 0.61–0.76 W across the 5 backbones at both FP32 and INT8, with no consistent precision-dependent difference. We do not report the corresponding measurement on the Raspberry Pi 4: the stock board exposes no on-board power or current sensor to software (`vcgencmd pmic_read_adc` returns no registered rail on this firmware), so board-level power is reported for the ZCU104 only.

The measured latency ranking (fastest to slowest: ShallowConvNet, EEGNet, DeepConvNet, Compact-CNN, FB-tCNN) is the same on both boards but does not match the ranking predicted from each backbone's maximum temporal kernel (Table 1) or from the floating-point operation count (FLOPs) measured directly to replace that proxy: only 2/5 positions match by kernel and 3/5 positions match by FLOPs, both below the majority threshold (4/5) set before measurement, a coarse test given only 5 possible rank positions. The discrepancy is concentrated in the two fastest backbones: EEGNet has the lowest FLOPs (0.87 million FLOPs at montage 3C) due to depthwise/separable convolution but was measured slightly slower than ShallowConvNet (4.03 million FLOPs) on both boards, showing that at this very small model scale, execution cost on an ARM CPU is not proportional to parameter count, kernel size, or theoretical FLOPs.

We synthesize EEGNet (montage 1C, BETA split 2026/seed 2026, CCKD student $\lambda_{sp} = 100$, 10-bit width) onto the ZCU104's PL. An initial attempt used hls4ml's VivadoAccelerator backend (AXI-Stream and AXI DMA); this reached a working synthesis and bitstream but a persistent TLAST-handling fault on the output stream, which survived six rounds of fixes to that backend, blocked on-board use. We therefore wrap the same already-verified inference kernel in a hand-written AXI memory-mapped (m_axi) interface with an AXI-Lite control register, avoiding the AXI-Stream path entirely; Vitis HLS and Vivado 2024.2 synthesis, implementation (place and route), and bitstream generation all succeed at a 100 MHz clock, with timing closing at a 2.074 ns margin. Whole-design resource usage (the Zynq processing system and the inference IP) uses 11.75% LUT, 6.54% FF, and 9.13% of Block RAM tiles of the XCZU7EV board, with no DSP blocks used, far below the 70% threshold locked in Section 4. We deploy this bitstream to the ZCU104 and run inference on the full 3200-window BETA test set for this setting: on-board accuracy is 24.06%, versus 24.09% for the H2 bit-accurate simulation at the same 10-bit width, a difference of 0.03 pp, well within the 1 pp threshold locked in Section 4. Board power during continuous inference (10.45–10.50 W) does not measurably exceed the idle baseline (10.55 W) at the resolution of the on-board INA226 sensor, consistent with the low utilization of this PL design.

5. Conclusion

CCKD does not provide a consistent benefit when extended from FB-tCNN to four other backbones: of the 20 backbone–dataset–montage settings, only 2 remained statistically significant after correction for multiple comparisons, and both were degradations (RQ1). The benefit of CCKD, when present, is sensitive to the specific montage chosen, not only to electrode count (RQ1b). The CKA increase between teacher and student caused by distillation predicts the accuracy gain, but the CKA gap before distillation does not; distilling the intermediate layer, alone or combined with the last layer, usually outperforms the last-layer distillation protocol [14] (RQ2). Evaluation on a single subject split, the protocol common in the EEG knowledge-distillation literature, misjudges the sign or statistical significance of the effect in half of the settings (RQ3). The CCKD student model keeps the sign of its benefit after INT8 quantization at the settings that remained statistically significant, tolerates fixed-point quantization down to 8–10 bits for most backbones tested, and every backbone achieves compute-only inference latency far below an illustrative wheelchair control-loop budget even at the lowest arithmetic width measured, excluding the SSVEP stimulation window itself; however, the measured latency does not rank by maximum temporal kernel or by FLOPs for most backbones, so selecting a backbone for a specific embedded platform requires direct measurement rather than inference from

the architecture (RQ4). The smallest backbone (EEGNet) synthesizes onto the ZCU104's PL within the board's resource budget (under 12% LUT, 7% FF, 10% BRAM, no DSP used) with timing closed, and its on-board inference accuracy matches the H2 bit-accurate simulation within 0.03 pp, well inside the 1 pp threshold. Overall, these results show that cross-montage relational distillation is backbone- and electrode-position-dependent, and that the sub-1–2-pp gains typical of this literature need to be evaluated over multiple subject splits with statistical correction, rather than stopping at a single split.

Reproducibility. The training, preprocessing, and statistical-analysis code, the per-split subject-assignment lists, the seeds, and the per-run results are publicly available at https://github.com/nguyenphuonghuy/SSVEP_CCKD. The software environment is version-pinned. Because GPU training is not bit-exact reproducible, repeated runs vary within the range measured in Section 4.4 (RQ3).

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